FLE TRIBE

WEEK 01 · WEDNESDAY · LESSON GUIDE

What is a browser?

This week: Explain step by step what happens between typing a URL and seeing a page.

The parts of a browser

- **User interface**: the address bar, tabs and buttons.
- **Networking**: downloads HTML, CSS, JavaScript and images.
- **Rendering engine**: turns HTML and CSS into pixels (Blink in Chrome and Edge, Gecko in Firefox, WebKit in Safari).
- **JavaScript engine**: runs your scripts (V8 in Chrome, SpiderMonkey in Firefox).

From code to pixels: the critical rendering path

- **1. Parse HTML → DOM**: the browser reads the HTML and builds a tree of elements.
- **2. Parse CSS → CSSOM**: it reads every stylesheet and builds a tree of styles.
- **3. Render tree**: DOM + CSSOM combined, keeping only what is visible.
- **4. Layout**: works out the exact size and position of every element.
- **5. Paint**: fills in pixels: text, colours, borders, images.
- **6. Composite**: layers are put together and shown on screen.

Why sites can look different in different browsers

Each engine is built by a different team and adds new features at its own pace. Check a feature on **caniuse.com** before relying on it, and always test your work in at least two browsers.

Research questions

Answer these in the app, in your own words, to unlock the next lesson.

1. Why can a site look different in Safari and Chrome?
2. What is the critical rendering path?

Start your research here

- MDN: https://developer.mozilla.org/en-US/docs/Web/Performance/Guides/How_browsers_work